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1. Coulomb's Law and the Electric Field

If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words? I believe it is the **atomic hypothesis** (or the atomic fact, or whatever you wish to call it) that *all things are made of atoms – little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another.*

– R. Feynman

All materials are made of atoms, and each atom is composed of a nucleus and surrounding electrons. Electrons carry negative charges, while the nucleus carries positive charges due to the positively charged protons. A neutral atom contains equal numbers of electrons and protons, so that there is no “net charge”. These charged particles interact with one another, and the *electromagnetic* interactions lead to rich and diverse phenomena. In fact, in our daily lives, the electromagnetic interactions may be the most important one among the four fundamental interactions.

1.1 Electric Charge

We have already mentioned that there are two types of electric charge, and Benjamin Franklin has named them **positive** and **negative**. It is well-known for all of you that the particles with opposite charge attract each other, whereas the particles with the same sign of charge repel each other. The *net charge* of an object is therefore defined as the total positive charge minus the magnitude of the total negative charge inside that object, or simply the algebraic sum of all the charges it contains. If the net charge of an object is zero, we call the object “charge neutral”.

In addition, electric charge is **quantized**. That means all charge is an integer multiple of a universal constant, or in a mathematical expression $q = ne$ ($n \in \mathbb{Z}$ is an integer)¹. Here e is called the elementary charge with a value of $|e| = 1.602 \times 10^{-19}$ (C). A proton has a positive elementary charge $|e|$, whereas an electron has a negative elementary charge $-|e|$. Since charge is carried by particles, it can be transferred from one body to another. It can also be created or annihilated, but equal amounts of positive charge and negative charge must be created or annihilated simultaneously. This is known as the **conservation of charge**. For example, one can annihilate an electron–positron pair (the positron carries positive elementary charge $|e|$) and create two γ -ray photons (which carry no charge).

Finally, there is one more important property of electric charge. The electric charge of an object is **Lorentz invariant**. This is in sharp contrast to the (relativistic) mass, which changes according to the speed of the object (recall relativity).

1.2 Coulomb's Law

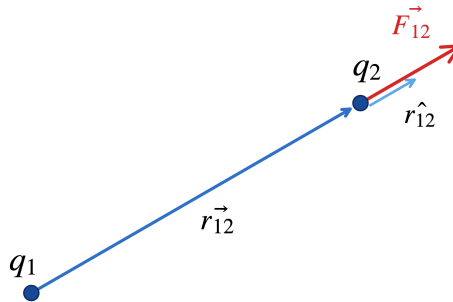


Figure 1.1: Coulomb force between two charged particles.

You are already familiar with Coulomb's law from high school physics. You may have the following equation in mind:

$$F = K \frac{qQ}{r^2}$$

However, this is not complete. We must keep in mind that the force is a vector so the direction of the force also matters.

In college general physics, we shall always write down the vector form of this formula, which states:

$$\mathbf{F}_{12} = K \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

Here, the subscript of $\mathbf{F}_{12}$ simply means the force exerted on particle 2 by particle 1; r_{12} means the distance between particle 1 and particle 2, whereas $\hat{r}_{12}$ is the unit vector from particle 1 to particle 2. You can verify that such a vector expression is consistent with all positive or negative charge cases.

¹Quarks carry $\pm e/3$ and $\pm 2e/3$, but are never observed in isolation.

In addition, to simplify the future equations (Maxwell's equations), we also use $\frac{1}{4\pi\epsilon_0}$ instead of K . Here $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$ is called *permittivity constant*. Now Coulomb's law can be written as:

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{\mathbf{r}}_{12} \quad (1.1)$$

The above equation applies to two static point charges. A point charge is an idealization: a charge of no physical size, or one whose dimensions are negligible compared with the distances over which the field is being calculated. If there are N static point charges, we can calculate the force between each pair of them using above equation. The total force on the i -th point charge $\mathbf{F}_i$ is therefore the vector sum of the force from all the rest particles. Therefore:

$$\mathbf{F}_i = \frac{1}{4\pi\epsilon_0} \sum_{j \neq i} \frac{q_j q_i}{r_{ji}^2} \hat{\mathbf{r}}_{ji}$$

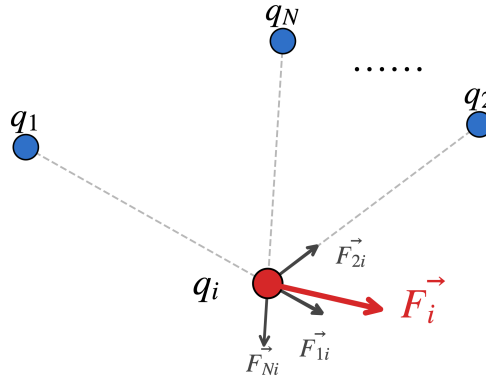


Figure 1.2: The total force on a point charge q_i due to multiple source charges q_j is the vector sum of individual Coulomb forces. This is the principle of superposition.

1.3 The Electric Field

Coulomb's law states that even though point charge 1 is not in contact with point charge 2, there is still an electrostatic force (Coulomb force) on particle 2 from particle 1. Perhaps you have asked this question: How can an object exert a force on another object at a distance without touching each other? Congratulations, you have asked a question that in principle leads eventually to the discovery of relativity. Faraday, Maxwell, Einstein, and all modern physicists believe that the charge carried by object 1 sets up an **electric field** in the surrounding space, which spreads across space-time. When we put object 2 in place, it is affected by the electric field at that specific point set up by the first object, and thus object 2 feels a force from object 1.

At first glance this may seem quite unnecessary, but I shall encourage you to imagine a case where both objects are moving in space, and keep in mind that the spreading of the electric field also takes time. You may probably see the difference.

Mathematically, the **field** is simply a space-time function that represents the distribution of a certain quantity. Here, the definition of electric field is according to the following: assuming that a small test charge q_0 is placed at $\mathbf{r}$ around a collection of charges, the electric field at this specific point $\mathbf{E}(\mathbf{r})$ is defined as:

$$\mathbf{E}(\mathbf{r}) = \mathbf{F}/q_0$$

Question: What do we mean “small”? Do we mean small amount of charge? Or do we mean the physical size of the test charge?

Using above definition, it is straightforward to show that the electric field due to a point charge q at the origin can be written as:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \quad (1.2)$$

where $\mathbf{r}$ is the position vector from the origin.

Similarly, for a set of point charges $q_1, q_2 \dots q_N$ at $\mathbf{r}_1, \mathbf{r}_2 \dots \mathbf{r}_N$, respectively, the electric field at $\mathbf{r}$ is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{|\mathbf{r} - \mathbf{r}_i|^3} (\mathbf{r} - \mathbf{r}_i) \quad (1.3)$$

We can use electric field lines to visualize the electric field and in many cases to help us understand the concept. There are several key points of electric field lines:

1. Electric field lines originate from positive charges and terminate at negative charges.
2. At any point, the tangent direction of the electric field lines represents the direction of the electric field at that specific point. Therefore, two field lines can never cross.
3. In the plane perpendicular to the electric field lines, the density of the lines at a specific point is proportional to the magnitude of the electric field at that point.

1.4 Electric Dipole

Consider a special charge system consisting of a pair of point charges separated by a small distance d , the two point charges having the same magnitude of charge q but opposite signs. An electric dipole is an ideal system with $d \rightarrow 0$ while $p = qd$ is kept constant. If we define $\mathbf{d}$ as the vector from the negative point charge to the positive point charge, the *dipole moment* $\mathbf{p}$ of the electric dipole is defined as $\mathbf{p} = q\mathbf{d}$. Below, we try to derive a general formula for the electric field of a dipole system. We shall assume that the center of the dipole is located at the origin, and we derive the expression for the electric field at position $\mathbf{r}$ using vector algebra. Don't panic, it's actually much simpler than you think.

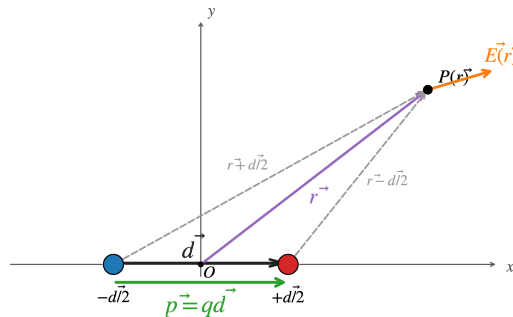


Figure 1.3: An electric dipole consists of two equal and opposite charges $\pm q$ separated by a distance d . The dipole moment is $\mathbf{p} = q\mathbf{d}$ where $\mathbf{d}$ points from $-q$ to $+q$.

Using above definition, this is simply a two-particle system with $q_1 = q$ located at $\mathbf{d}/2$ and $q_2 = -q$ located at $-\mathbf{d}/2$. Thus

$$\begin{aligned}\mathbf{E}(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{|\mathbf{r} - \mathbf{r}_i|^3} (\mathbf{r} - \mathbf{r}_i) \\ &= \frac{1}{4\pi\epsilon_0} q \left(\frac{\mathbf{r} - \frac{\mathbf{d}}{2}}{|\mathbf{r} - \frac{\mathbf{d}}{2}|^3} - \frac{\mathbf{r} + \frac{\mathbf{d}}{2}}{|\mathbf{r} + \frac{\mathbf{d}}{2}|^3} \right)\end{aligned}$$

We note that

$$|\mathbf{r} - \frac{\mathbf{d}}{2}|^{-3} = (r^2 - \mathbf{r} \cdot \mathbf{d} + \frac{d^2}{4})^{-3/2} \approx r^{-3} (1 + \frac{3}{2} \frac{\mathbf{r} \cdot \mathbf{d}}{r^2})$$

when $d \ll r$. Similarly

$$|\mathbf{r} + \frac{\mathbf{d}}{2}|^{-3} = (r^2 + \mathbf{r} \cdot \mathbf{d} + \frac{d^2}{4})^{-3/2} \approx r^{-3} (1 - \frac{3}{2} \frac{\mathbf{r} \cdot \mathbf{d}}{r^2})$$

Substituting them into the original expression, we have:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$$

It is fairly simple to prove that such an electric dipole system experiences no net force in a uniform electric field. However, there will be generally a torque $\boldsymbol{\tau}$ on the dipole system:

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$$

1.5 Continuous Charge

Equations introduced above deal with discrete point charge systems. Nevertheless, they can be immediately extended to “continuously” distributed charge systems. The idea is simply slicing the charge system into many tiny pieces, each of which can be regarded as a “point charge”, and summing up their contributions. When this is taken to the infinite limit, it becomes an integral. In this spirit, Eq. 1.1 becomes:

$$\mathbf{F}_0 = \frac{1}{4\pi\epsilon_0} \int_q \frac{q_0 dq}{r_q^2} \hat{\mathbf{r}}_q \quad (1.4)$$

Note that the above equation means the force on a point charge q_0 (which is not part of the continuous charge system q); r_q is the distance from the infinitesimal charge dq to q_0 , $\hat{\mathbf{r}}_q$ is the corresponding unit vector; and $\int_q$ means the integral takes all “sliced” pieces into consideration.

Question: We just mentioned that charges are “quantized”. So what do we mean by “continuous” charge and “integral”?

Similarly, the electric field due to a continuously distributed charge system q :

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_q \frac{dq}{r_q^2} \hat{\mathbf{r}}_q \quad (1.5)$$

Please take some time to think about the difference between the equation 1.5 and 1.3, especially the relation between r_q and $\mathbf{r} - \mathbf{r}_i$. In many cases, the mathematical expression in physics should not be taken “literally”.

Example 1.1: Electric field due to charged disk.

We have a circular disk of radius R uniformly charged with Q . Calculate the electric field at P located on the axis of the disk and z away from its center.

Solution: We take the axis of the disk as $\hat{z}$, and its center at the origin. The charge element on the disk is taken by first slicing the disk into rings from r to $r + dr$, and then angularly from ϕ to $\phi + d\phi$. Therefore, the surface area element is $dS = r dr d\phi$. The charge is uniformly distributed, thus the charge element is:

$$dq = \sigma r dr d\phi$$

where $\sigma = \frac{Q}{\pi R^2}$ is the charge per unit area (surface charge density) and its distance to P is $\sqrt{z^2 + r^2}$. Noticing that the system is axially symmetric, meaning that components perpendicular to z will cancel each other, therefore we only need to calculate the $\hat{z}$ component. Thus

$$\begin{aligned} dE_z &= \frac{1}{4\pi\epsilon_0} \frac{dq}{r_q^2} \hat{r}_q \cdot \hat{z} \\ &= \frac{1}{4\pi\epsilon_0} \frac{\sigma z r dr d\phi}{(z^2 + r^2)^{3/2}} \end{aligned}$$

The final answer is obtained by complete integration:

$$\begin{aligned} E_z &= \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r_q^2} \hat{r}_q \cdot \hat{z} \\ &= \frac{\sigma z}{4\pi\epsilon_0} \int_0^R dr \int_0^{2\pi} d\phi \frac{r}{(z^2 + r^2)^{3/2}} \\ &= \frac{\sigma z}{4\pi\epsilon_0} \int_0^R dr \frac{2\pi r}{(z^2 + r^2)^{3/2}} \\ &= \frac{zQ}{2\epsilon_0\pi R^2} \left(\frac{1}{z} - \frac{1}{\sqrt{z^2 + R^2}} \right) \end{aligned}$$

We took $z > 0$ in the above solution; and the field for $z < 0$ follows from the mirror symmetry of the disk.

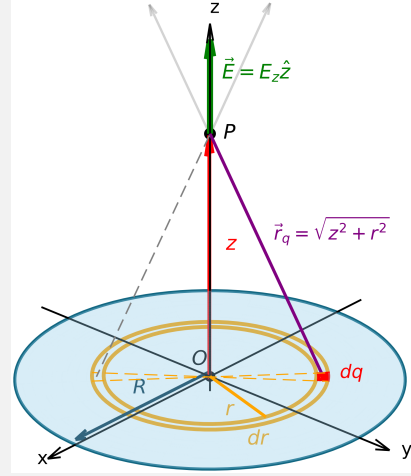


Figure 1.4: A uniformly charged disk of radius R . The disk is divided into concentric rings (r to $r + dr$). Due to axial symmetry, only the z -component dE_z contributes at point P on the axis.

1.6 Exercises

1. An electric quadrupole can be viewed as 4 point charges at 4 corners of a square of side a , a pair of $+q$ located at one of the diagonals, and a pair of $-q$ located at the other. Suppose a is small but $\Omega = qa^2$ is finite. Use Ω to express the electric field generated by this quadrupole at P :
 - (a) P is located along the diagonal of $-qs$, and at distance x away from the center of the square.
 - (b) P is located along the axis perpendicular to one of the edges and through the center, and at distance x away from the center of the square.
2. Explicitly use $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$ to calculate the electric field generated by a *uniformly* charged spherical shell of radius R carrying total charge Q , both inside ($r < R$) and outside ($r > R$). The thickness of the shell can be ignored.
3. An electric dipole $\mathbf{p}_2$ is located at position $\mathbf{r}$ relative to a second dipole $\mathbf{p}_1$, with r much larger than the size of either dipole. Find the force exerted on $\mathbf{p}_2$ by $\mathbf{p}_1$, expressed in terms of $\mathbf{p}_1$, $\mathbf{p}_2$ and $\mathbf{r}$. (Hint: write $\mathbf{p}_2$ as a pair of charges $\pm q_2$ at $\mathbf{r} \pm \mathbf{d}_2/2$, and expand the field of $\mathbf{p}_1$ to first order in $\mathbf{d}_2$, as was done for the dipole field itself in Section 1.4.)
4. Explicitly use $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$ to calculate the electric field generated by a uniformly charged solid sphere of radius R carrying total charge Q , both inside ($r < R$) and outside ($r > R$).
5. Calculate the electric field generated by a solid cylinder of radius R and length L , carrying a total charge Q uniformly distributed throughout its volume, at P :
 - (a) P is located on the axis of the cylinder, at distance $x > L/2$ from its center.
 - (b) P is located in the mid-plane perpendicular to the axis of the cylinder, at distance $x \gg R$ from the axis.
6. Four identical point charges $q > 0$ are held at the four corners of a square of side a .
 - (a) Find the magnitude and the direction of the net electrostatic force on any one of them.
 - (b) A fifth point charge Q is now placed at the center of the square. Find the value of Q for which all five charges are in equilibrium.
7. A thin rod occupies the half-line $z \geq 0$ and carries a uniform linear charge density $\lambda > 0$; such a rod is called semi-infinite. The point P lies in the plane $z = 0$, at a perpendicular distance d from the end of the rod.
 - (a) Calculate both the component of $\mathbf{E}$ perpendicular to the rod and the component parallel to it at P , taking $\hat{z}$ as positive for the parallel one, and show that $\mathbf{E}$ makes an angle of 45° with the axis of the rod no matter how large or small d is.
 - (b) Without evaluating any new integral, use the result of (a) to obtain the electric field at a perpendicular distance d from an infinitely long rod carrying the same λ .
8. Calculate the electric field at the center of curvature of the following two charge distributions.
 - (a) A thin circular arc of radius R carrying a total charge $Q > 0$ spread uniformly along its length, subtending an angle 2α at the center, where $0 < \alpha \leq \pi$. Check your answer in the two limits $\alpha \rightarrow 0$ and $\alpha = \pi$.

- (b) A full circular ring of radius R , lying in the xy plane and centered at the origin, carrying the non-uniform linear charge density $\lambda(\phi) = \lambda_0 \cos \phi$ with $\lambda_0 > 0$, where ϕ is the azimuthal angle measured from the positive x axis. What is the total charge on this ring?
9. A rigid electric dipole is made of two point charges $+q$ and $-q$, each of mass m , fixed at the two ends of a massless rod of length d , so that its dipole moment has magnitude $p = qd$.
- (a) The dipole is placed in a uniform electric field $\mathbf{E}$ and released from rest at a small angle from the direction of $\mathbf{E}$. Show that it rotates about its center and oscillates harmonically about the direction of $\mathbf{E}$, and find the angular frequency of the oscillation.
- (b) The uniform field is now replaced by the field of a point charge $Q > 0$ fixed at the origin. The dipole sits on the z axis at $z = r \gg d$, with $\mathbf{p} = p\hat{z}$, that is, pointing away from Q . Find the net force on the dipole to leading order in d/r , and verify that it is equal and opposite to the force that the dipole exerts on Q .
10. The atoms of Feynman's quote are held together by the electric force, and they are neutral to an astonishing accuracy. This exercise estimates both statements. Use $e = 1.60 \times 10^{-19}$ C, $\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9$ N·m²/C², $G = 6.67 \times 10^{-11}$ N·m²/kg², $m_e = 9.11 \times 10^{-31}$ kg, $m_p = 1.673 \times 10^{-27}$ kg and $m_H = 1.674 \times 10^{-27}$ kg.
- (a) Calculate the ratio of the electrostatic force to the gravitational force between one electron and one proton, and explain why this ratio does not depend on how far apart they are.
- (b) Suppose the two charges did not cancel exactly, the electron carrying $-e(1 + \delta)$ while the proton carries $+e$, so that a hydrogen atom would carry a net charge $-\delta e$. Find the value of $|\delta|$ for which the electrostatic repulsion between two hydrogen atoms would exactly cancel their gravitational attraction, and show that this value is also independent of their separation.